

Data Scientist Assignment: Candidate Profiling via Clustering

Objective

We have **thousands of candidate assessments**, each containing:

- A list of **hard skills** extracted by an LLM from transcript
- A **score** for each skill
- A full **interview transcript**

Your task is to design an approach that clusters candidates into **meaningful profiles** based on skills, scores, and interview content (if needed).

Important Requirements

1. Hard Skills Are LLM-Generated and Noisy

You must account for:

- High variation in naming (e.g., *Python*, *Python Programming*, *Machine Learning Modeling*, Machine Learning, Postgre, PostgreSQL, SQL, etc.)
- Synonyms
- Different levels of specificity (Python, pandas, scikit-learn, pytorch,...)
- Up to several thousand unique raw skill labels
- Inconsistent formatting and potential typos

Your solution might consider a way to **normalize**, **standardize**, and **represent** these skills effectively.

2. Skill Relationship Mapping

Your approach must be able to model **relationships between skills**, including:

- Skill relationship mapping
 - e.g., if someone knows *Next.js*, they should implicitly be represented as knowing *JavaScript*

- Broader domain or category relationships
- The ability to infer or assign skill dependencies

This requirement must be clearly addressed in your design.

3. Clustering Requirement

Your clustering approach must:

- Use a combined representation of skills, inferred skills, scores, and interview transcript features
- Produce interpretable candidate profiles (why they are belong to this group)
- Also include the skills that they are not good at in the clustering
- Group label for a group:
 - *“Strong Docker + strong data validation + weak Kubernetes HA.”*
 - *“Great at schema thinking, weaker at failover architecture.”*
 - *“Well-rounded but shallow vs specialized but deep.”*
- Scale to thousands of assessments and thousands of skills

You must justify:

- What features you use
- How you represent candidates
- How you determine the number and quality of clusters
- How you ensure clusters are meaningful and actionable

Dataset Description

Each record includes:

- `candidate_id`
- `skills`: raw LLM-generated skill list from transcript
- `skill_scores`: numeric score per skill
- `interview_transcript`: free text
- (Optional) metadata such as experience level or applied role

Your Tasks

1. Propose a complete end-to-end approach
2. Provide effort estimation
 - A breakdown of tasks
 - Estimated time needed

- Potential risks or unresolved challenges
- Any assumptions you made

3. Due date: 1 day